

KESHAV GUPTA

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EDUCATION

Indian Institute of Information Technology, Kota

2022 – 2026

Bachelor of Technology in Electronics and Communication Engineering [ECE]

WORK EXPERIENCE

AI ML Intern – Edunet Foundation (Microsoft-AICTE Program)

Apr 2025 – May 2025

- Engineered an end-to-end NLP application by designing a data pipeline for text extraction, preprocessing, and performing sentiment analysis using Hugging Face transformer models.
- Conducted statistical analysis on chat logs, performing data modeling to identify key engagement drivers and visualize communication patterns for stakeholder review via a Streamlit dashboard.

PROJECTS

Whisper-Small Fine-Tuning for Hindi ASR (Deep Learning)

- Improved ASR accuracy by 50.1%** (reducing WER to 42.01%) by designing a complete ETL pipeline and fine-tuning the 'openai/whisper-small' model on a custom Hindi dataset.
- Accelerated model training and inference on GPU hardware using Hugging Face 'transformers' and mixed-precision (FP16), preparing the model for production-level deployment.

Customer Churn Prediction (Predictive Modeling)

- Developed a predictive modeling framework that achieved 79% accuracy and 64% recall**, providing a reliable tool for identifying at-risk customers from a large-scale Telco dataset.
- Executed comprehensive feature engineering and selection, then applied SMOTE to rectify significant class imbalance, boosting model performance for business intelligence applications.
- Optimized machine learning models (XGBoost, LightGBM) using hyperparameter tuning (GridSearchCV) and validated model performance using a suite of metrics including ROC/AUC curves.

Twitter Sentiment Analysis (Natural Language Processing)

- Made a Transformer-based deep learning model** for large-scale text classification, achieving 81.37% test accuracy on a 1.6M tweet dataset.
- Implemented regularization techniques including early stopping and learning rate scheduling to mitigate overfitting and improve model generalization on unseen data.
- Performed statistical analysis and model comparison between RNN (LSTM) and Transformer architectures to inform model selection for large-scale NLP tasks.

TECHNICAL SKILLS

Languages & Databases: Python, C++, C, SQL, JavaScript, HTML/CSS

ML Libraries Frameworks: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Streamlit, LightGBM, XGBoost, Hugging Face Transformers

Tools, Platforms & Cloud: Git, GitHub, Docker, CI/CD, Azure, AWS (S3, EC2), Google Colab, Jupyter Notebook, REST APIs

ML Concepts: Data Science, Data Analysis, Data Visualization, ETL Pipelines, Predictive Modeling, NLP, ASR, Computer Vision, Generative AI, Supervised Unsupervised Learning, Classification, Regression, Statistical Analysis, Feature Engineering, Hyperparameter Tuning, Model Evaluation Validation

CERTIFICATIONS

- Machine Learning Specialization by Andrew Ng - Coursera
- Generative AI Prompt Engineering (24+ hours) - Udemy

LEADERSHIP EXTRACURRICULAR

Logistics Secretary, CACS Creative Arts and Cultural Society, IIIT Kota

- Led a 30-member team to manage logistics and operations for cultural events with over 500 attendees.

Social Media Lead Event Manager, Incognito, Kernel, Fit India Clubs

- Managed promotions and digital strategies across three clubs, boosting student engagement by 25%.